Reasoning as Infrastructure

A Compositional Architecture for Regulatory AI Alignment in Organ Procurement Without Fine-Tuning

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Abstract

Organ procurement organizations face mounting pressure to adopt artificial intelligence for regulatory compliance under the Centers for Medicare and Medicaid Services final rule CMS-3409-P, which establishes a three-tier performance evaluation system with decertification consequences. Yet no established methodology exists for aligning general-purpose large language models with the complex, multi-authority regulatory environment governing organ procurement. Standard retrieval-augmented generation addresses the knowledge retrieval problem but fails to address what this paper identifies as the *reasoning pattern gap*: AI systems consistently fail not because they lack the relevant regulatory text, but because they navigate multi-authority intersections incorrectly—misattributing jurisdiction, conflating temporal status, collapsing regulatory hierarchy, or generating confident answers where genuine gaps exist.

This paper presents a compositional architecture that treats regulatory reasoning as *infrastructure*—a manufacturable, versionable, portable artifact that can be deployed to any frontier language model through the context window, without fine-tuning or weight modification. The framework decomposes domain reasoning into typed, composable primitives organized by a formal grammar; routes authority identification and temporal resolution deterministically before the language model engages; enforces safety boundaries through automaton topology rather than textual instruction; and extends beyond reactive question-answering to proactive regulatory co-navigation through obligation tracking and decision cascade mapping.

The architecture is organized in seven layers: a deterministic pre-processing substrate, a compositional reasoning grammar, a safety automaton, an inference engine, an audit-grade output contract, a continuous evolution loop, and a proactive navigation layer. The system prompt consumes approximately five thousand tokens. The compositional grammar generates correct reasoning chains for regulatory intersections the system has never encountered. The safety guarantee is structural: the Human Line—the boundary between AI-assisted guidance and decisions that must remain with human practitioners—is enforced by the topology of the reasoning automaton, not by prompt engineering.

The framework is portable across models, organizations, and regulatory updates. A shared community library of archetypal reasoning primitives enables all fifty-six OPOs to benefit from a common reasoning infrastructure while maintaining organization-specific customization. All specifications are released under open license to support industry-wide adoption.

Keywords: *organ procurement, regulatory compliance, compositional reasoning, retrieval-augmented generation, in-context learning, domain alignment, CMS-3409-P, OPTN, OPO, decision support, safety automaton*

1. Introduction

At 3:14 AM, a transplant coordinator receives notification of a potential organ donor. Within the next several hours, she must navigate a regulatory environment spanning federal statute (the National Organ Transplant Act), CMS conditions of participation (42 CFR 486 Subpart G), Organ Procurement and Transplantation Network policies and bylaws, her state's Uniform Anatomical Gift Act, and her organization's internal standard operating procedures. Many of these authorities address overlapping domains. Some conflict. The consequences of error range from regulatory citation to organ loss to patient death.

She does not need an AI system that retrieves the right policy text. She needs an AI system that *reasons correctly about how multiple regulatory authorities intersect* in the specific operational scenario she faces—and that knows what it must not attempt to decide.

The fifty-six organ procurement organizations operating in the United States are under escalating pressure to adopt AI-assisted decision support. CMS final rule CMS-3409-P, published January 30, 2026, establishes outcome-based performance metrics with a three-tier evaluation system. OPOs failing to meet benchmarks face decertification within approximately eighteen months. Simultaneously, the transplant community recognizes that AI tools could meaningfully improve organ utilization, allocation compliance, and documentation quality. However, no established framework exists for ensuring that AI systems reason correctly about OPO regulatory compliance—as distinct from merely retrieving relevant policy text.

This paper addresses this gap with a compositional architecture built on a specific thesis: **regulatory reasoning competence is a separable, storable, transferable artifact that exists independently of the model that produced it and the model that consumes it.** It is a third category of AI asset—not weights, not data, but crystallized methodology that can be manufactured once and deployed infinitely.

The architecture decomposes multi-authority regulatory reasoning into typed, composable primitives—atomic reasoning operations analogous to functions in a programming language—organized by a formal composition grammar. Where traditional approaches store complete reasoning demonstrations (analogous to storing complete sentences), this framework stores the grammar from which correct reasoning chains are generated on demand (analogous to storing the rules of a language). A small set of primitives composes into reasoning chains for regulatory intersections the system has never encountered, including intersections that did not exist when the primitive library was built.

Critically, the architecture recognizes that the highest-value mode for organ procurement is not answering regulatory questions—it is preventing the coordinator from missing regulatory obligations she did not know to ask about. The framework extends beyond reactive question-answering to proactive regulatory co-navigation: tracking obligations across the case lifecycle, surfacing downstream consequences of decisions before they are made, and providing stage-transition briefings as cases move through the procurement process.

The contributions of this paper are as follows. First, we identify the reasoning pattern gap as distinct from the knowledge retrieval gap in regulated healthcare AI, and catalog its characteristic failure modes. Second, we present a seven-layer compositional architecture that routes authority identification deterministically, composes reasoning from typed primitives, enforces safety boundaries structurally, produces audit-grade outputs, evolves through operational feedback, and provides proactive regulatory navigation. Third, we specify the complete framework—primitive format, composition grammar, automaton specification, generation pipeline, and deployment architecture—in sufficient detail for implementation by OPO information technology teams using current infrastructure. All specifications are released under open license.

2. The OPO Regulatory Environment

Organ procurement organizations operate under a multi-layered regulatory framework with a finite but interacting set of authoritative sources. At the statutory level, the National Organ Transplant Act (NOTA, 42 U.S.C. § 274) establishes the legal foundation for organ procurement and the OPTN. At the regulatory level, CMS conditions of participation (42 CFR 486 Subpart G) define certification requirements that OPOs must meet to receive Medicare reimbursement. The OPTN, operated under contract with the Health Resources and Services Administration, promulgates policies governing organ allocation, distribution, and member conduct. State Uniform Anatomical Gift Acts add jurisdiction-specific requirements for organ donation authorization. Individual OPOs maintain internal standard operating procedures that implement these overlapping requirements.

The regulatory landscape is not static. CMS-3409-P introduces a new outcome-based performance evaluation framework that substantially changes how OPOs are measured, evaluated, and potentially decertified. OPTN policies undergo regular revision through a public comment process. State legislatures periodically update UAGA provisions. An AI system operating in this environment must not only know the current regulatory state but reason about temporal dimensions: which rules are current, which are proposed, which are pending finalization, and how pending changes alter the regulatory hierarchy.

CMS-3409-P creates a specific and urgent complication. Under the new framework, regulatory *compliance* and regulatory *optimization* are no longer synonymous. A coordinator can be fully compliant with OPTN allocation policy while making decisions that damage the OPO’s CMS performance tier. The new rule has a phased implementation timeline—some provisions take effect immediately, others have delayed effective dates, and transition periods apply to certain metrics. An AI system must model these phases, not merely classify rules as “current” or “proposed.”

2.1 Retrieval-Augmented Generation and Its Limitations

Retrieval-augmented generation (RAG) has emerged as the standard approach for grounding language model responses in authoritative source documents. In a RAG architecture, user queries are embedded and matched against a vector index of document chunks; the most relevant chunks are injected into the model’s context alongside the query. RAG effectively addresses the knowledge retrieval problem: given a question about OPTN Policy 2.6, the system retrieves the relevant policy sections and provides an accurate response.

However, RAG does not address the reasoning pattern problem. When a query requires synthesizing information across multiple regulatory authorities—determining jurisdictional primacy, resolving apparent conflicts, accounting for temporal status of proposed rules—the model must perform multi-hop reasoning that retrieved text chunks alone do not scaffold. The retrieval layer delivers the raw materials; the reasoning layer must construct the analysis. In regulated healthcare, errors in the reasoning layer carry consequences that errors in the retrieval layer do not.

2.2 Extended Test-Time Compute

Recent advances in language model inference have introduced extended test-time compute—systems that allocate substantially more computation per query to produce more thoroughly reasoned responses. These systems demonstrate marked improvements on complex reasoning tasks, including multi-step logical inference and cross-document synthesis. However, extended thinking modes require minutes rather than seconds, making them unsuitable for real-time interactive applications where a transplant coordinator needs guidance at 3 AM. This creates a fundamental tension: the highest-quality reasoning requires compute budgets incompatible with operational latency requirements.

The architecture presented in this paper resolves this tension by separating the *generation* of complex regulatory reasoning (performed offline with extended compute) from its *deployment* (performed at inference time through the context window at standard latency). The expensive reasoning is performed once; its products—validated reasoning primitives—are deployed infinitely at near-zero marginal cost.

3. The Reasoning Pattern Gap

We define the *reasoning pattern gap* as the systematic failure of retrieval-augmented language models to correctly navigate multi-authority regulatory intersections, even when all relevant source material is present in the context window. This gap manifests in five characteristic failure modes, plus a sixth introduced by CMS-3409-P.

3.1 Failure Mode Catalog

Jurisdictional misattribution. The model applies the wrong authority to a given scenario. For example, citing CMS conditions of participation for a question that falls under OPTN allocation policy, or vice versa. Both authorities may address related topics, but which governs depends on the specific operational context.

Temporal conflation. The model treats proposed rules as binding, or fails to distinguish between current regulations and pending amendments. CMS-3409-P is particularly vulnerable to this failure: as a final rule with a phased implementation timeline, its provisions occupy a complex temporal space that requires careful phase-aware reasoning.

Hierarchy collapse. The model fails to recognize or correctly apply the regulatory hierarchy (statute → regulation → OPTN policy → OPO internal procedure). When authorities appear to conflict, the model may defer to whichever was retrieved first, or confuse *specificity* with *superiority*—treating a more specific OPTN policy as overriding a less specific but higher-authority CMS regulation.

Gap blindness. The model fails to recognize genuine regulatory gaps—scenarios where no authority provides clear guidance. Instead of flagging the gap and recommending appropriate escalation, the model generates a confident answer by over-extending the scope of an adjacent provision.

Boundary violation. The model provides clinical determinations, eligibility assessments, or family communication recommendations that exceed the appropriate scope of AI-assisted decision support. These represent failures of domain-appropriate behavioral calibration.

Metric blindness. Introduced by CMS-3409-P: the model makes a technically correct regulatory determination but fails to account for how the decision affects OPO performance metrics under the new evaluation framework—or, conversely, allows metric optimization to override proper regulatory analysis.

3.2 Where Failures Cluster

These failure modes are not random. They cluster at regulatory intersections—the seams where two or more authorities overlap. A model reasoning within a single authority typically performs adequately. The failures emerge when the reasoning must span boundaries. An inverse relationship obtains between intersection frequency and failure consequence: multi-authority intersections are rare (estimated five to ten percent of coordinator queries), but they carry disproportionate risk because they represent exactly the scenarios where unscaffolded models fail most dangerously—often with high confidence.

This observation motivates the central architectural decision of this framework: to decompose regulatory reasoning into composable primitives that can be combined to handle *any* intersection—including intersections that have never been encountered before—rather than attempting to pre-compute a demonstration for every possible scenario.

4. Architecture Overview

The framework is organized in seven layers, each addressing a specific class of problem. The layers are independently valuable—an organization can implement Layer 1 alone and see immediate benefit—but they compose into a system whose capabilities exceed any individual layer.

Layer	Name	Function	Compute
1	Deterministic Substrate	Authority routing, temporal resolution, query canonicalization	No LLM required
2	Compositional Grammar	Typed reasoning primitives, counter-traces, composition rules	Pre-computed offline
3	Safety Automaton	Structural completeness verification and boundary enforcement	Prompt architecture
4	Inference Engine	Grammar compilation, triple retrieval, divergence calibration	Standard LLM latency
5	Output Contract	Evidence packs, cascade mapping, audit trail	Structured post-processing
6	Continuous Evolution	Feedback, harm-prioritized generation, diff propagation	Background/periodic
7	Proactive Navigation	Obligation tracking, stage briefings, consequence modeling	Mostly deterministic

Three design constraints are non-negotiable across all layers. **Portability:** the system operates through the context window and works with any language model that accepts system prompts, without fine-tuning or weight modification. **Implementability:** every component is buildable by OPO information technology teams using standard infrastructure (Python, vector databases, standard APIs). **The Human Line:** the boundary between AI-assisted guidance and decisions reserved for human practitioners is enforced structurally, not by textual instruction.

5. Layer 1: The Deterministic Substrate

The single highest-leverage architectural decision in this framework is to resolve authority identification, temporal status, and jurisdictional routing *before the language model engages*, using deterministic, non-LLM components. This eliminates the most common and most dangerous failure class—jurisdictional misattribution—without spending a single language model token.

5.1 Query Canonicalization

Transplant coordinators under operational pressure do not ask clean questions. A 3 AM query might arrive as "patient crashing, family weird, what do?" The system must not route this raw input directly into the reasoning engine.

A lightweight canonicalization component (a small rule-based system or dedicated small model, distinct from the reasoning language model) rewrites the coordinator's input into a structured canonical form:

> { intent: authorization_guidance, case_type: unknown, complicating_factors: [family_resistance], jurisdiction: TX, stage: approach, risk_flags: [boundary_risk] }

The canonicalizer outputs a constrained schema or fails closed—returning "unclear, needs clarification" rather than passing garbage input forward. This is defense in depth: the reasoning engine only ever receives well-formed queries.

5.2 The Authority Router

Authority identification in the OPO domain is not a reasoning problem. It is a classification problem. For a bounded domain with a finite set of authorities, a hand-built, domain-expert-reviewed decision structure maps the query's features (intent, case type, jurisdiction, workflow stage) to an authority set with a precedence ordering.

The authority router has three output modes to handle varying confidence levels:

Mode	Condition	Action
Deterministic	Router maps query to authority set with high confidence (~75% of queries)	Emit authority itinerary as pre-computed fact
Probabilistic	Router identifies 2–3 plausible authority sets (~20% of queries)	Emit all plausible itineraries, ranked; LLM instructed to reason through which applies
Escalate	Router cannot classify the query (~5% of queries)	Flag for LLM-assisted authority identification using the IDENTIFY primitive, with explicit uncertainty marking

The deterministic mode is the primary operational path. The model receives the authority itinerary as a fact—like it receives the date. The language model's job is limited to reasoning about intersections and synthesizing guidance: the task it performs well when properly scaffolded.

5.3 Temporal Phase Resolution

A maintained registry resolves the temporal status of each regulatory provision. Rather than the binary classification (current vs. proposed) that most systems employ, the resolver models regulatory phases:

> CMS-3409-P § 486.360(b): status = final_rule; transition phase from 2026-01-30 to 2027-07-01 (prior methodology applies, new metrics calculated but not determinative); full implementation from 2027-07-01 (three-tier evaluation fully operative).

Every query is anchored to a specific case date and time. The system stamps each response: "Regulatory state as of [date/time]. If incorrect, results may change." This converts temporal conflation from a reasoning failure into a data-entry problem.

5.4 Anticipatory Context Loading

When a case opens in the OPO's case management system, the router computes the likely authority profile for that case type and pre-loads the relevant reasoning primitives and policy chunks into a case-specific context cache. When the coordinator's first question arrives, the reasoning scaffold is already warm—zero retrieval latency, full contextual coherence.

As the case moves through stages (referral → evaluation → authorization → allocation → recovery → distribution), the active context rotates: authorization-stage primitives activate when the case enters authorization; allocation primitives activate at allocation. The system manages context throughout the case lifecycle, not just at case open.

Shift handoff continuity follows naturally: when a case transfers between coordinators, the case context transfers with it. The incoming coordinator inherits the full reasoning scaffold and case obligation state, not merely case notes.

5.5 The Authority Itinerary

Layer 1 outputs a structured Authority Itinerary that accompanies every query into the inference engine:

> { case_time_anchor: "2026-02-19T03:14:00-06:00", jurisdiction: { state: "TX", federal: true }, stage: "authorization", authority_itinerary: [{ id: "UAGA-TX", status: "current", scope:

"authorization", precedence: 1 }, { id: "OPTN-Policy-2.0", status: "current", scope: "allocation_authorization", precedence: 2 }, { id: "OPO-SOP-AUTH", status: "current", scope: "procedure", precedence: 3 }], intersection_signature: "UAGA+OPTN+SOP", predicted_complexity: "edge", risk_flags: ["temporal_risk", "conflict_risk"] }

This itinerary is injected as a pre-computed input. The language model does not determine which authorities apply; it reasons about the intersections between authorities it has been told apply. This separation of concerns is the foundation upon which every subsequent layer builds.

6. Layer 2: The Compositional Grammar

The core structural innovation of this framework is the decomposition of regulatory reasoning into typed, composable primitives organized by a formal grammar. Where traditional approaches store complete reasoning demonstrations—analogous to storing a library of solved problems—this framework stores the grammar from which correct reasoning chains are generated on demand.

6.1 Primitive Types

The OPO regulatory domain requires exactly eight types of reasoning primitive. This is not arbitrary: seven map directly to the failure modes identified in Section 3, and the eighth (CASCADE_MAP) enables the proactive navigation capabilities of Layer 7.

Primitive Type	Symbol	Addresses	Token Budget
HIERARCHY_RESOLVE	$H(a_1, a_2)$	Hierarchy collapse	150–300
TEMPORAL_CLASSIFY	T(provision, date)	Temporal conflation	100–200
CONFLICT_ADJUDICATE	$C(a_1, a_2, \text{type})$	Jurisdictional misattribution	200–400
GAP_DETECT	G(query, authorities)	Gap blindness	150–250
BOUNDARY_ENFORCE	B(query_intent)	Boundary violation	100–200
CASCADE_MAP	M(decision, authorities)	Metric blindness / downstream obligations	200–400
SYNTHESIZE	S(resolved[])	Final assembly of determination	200–400
CALIBRATE	K(chain, evidence)	Confidence assignment	100–150

Each primitive is stored in two forms. The **skeletal form** (three to five lines, always injected) captures the bare operational logic—what the primitive does, when to apply it, what it outputs. The **expanded form** (a full reasoning demonstration with worked examples and edge cases) is retrieved on demand when complexity warrants. This dual-form storage yields better model uptake per token than long, verbose demonstrations, directly addressing the attention dilution risk in large context windows.

6.2 Primitive Specification

Each primitive is a structured object with the following fields: a unique identifier encoding the primitive type and authority pair; an input signature specifying the authorities and conflict type; the skeletal and expanded reasoning forms; a paired counter-trace (Section 6.3); staleness atoms identifying which regulatory changes would invalidate the primitive; generation and validation metadata; and a composability specification indicating which primitives it requires before it and which it feeds into.

The expanded form of a HIERARCHY_RESOLVE primitive for the CMS↔OPTN pair illustrates the level of reasoning guidance:

> *Step 1: Identify the specific provisions from each authority that apply to this scenario. Step 2: Classify the relationship—does OPTN add requirements beyond CMS (additive)? Or does OPTN specify a different standard that conflicts with CMS (contradictory)? Step 3: If additive, OPTN requirements layer on top of CMS; the coordinator must meet both. Step 4: If contradictory, CMS governs; OPTN cannot override federal regulation. Step 5: Common trap—treating "OPTN is more specific" as "OPTN overrides CMS." Specificity does not equal superiority. The hierarchy is: statute > federal regulation > OPTN policy > OPO SOP. Specificity operates within a tier, not across tiers.*

6.3 Counter-Traces: The Immunization Layer

For each high-risk primitive, the library includes a paired **counter-trace**: a compressed demonstration of the *wrong* reasoning pattern paired with a forensic diagnosis of why it fails and what the correct primitive sequence is. Counter-traces exploit a well-established property of in-context learning: models calibrate substantially better when shown both positive and negative examples.

Counter-Trace	Failure Mode Addressed	Trap Pattern
CT-JURISDICTION	Jurisdictional misattribution	Applying CMS to an OPTN-governed question
CT-TEMPORAL	Temporal conflation	Citing a proposed rule as binding authority
CT-HIERARCHY	Hierarchy collapse	"More specific = higher authority"
CT-GAP	Gap blindness	Confident answer extrapolated from adjacent provision
CT-BOUNDARY	Boundary violation	Drifting into clinical determination

Counter-Trace	Failure Mode Addressed	Trap Pattern
CT-SCOPE-CREEP	Gradual boundary erosion	Three-question escalation toward forbidden territory
CT-METRIC-BLIND	Metric-driven reasoning	CMS-3409-P metrics overriding regulatory analysis

Total counter-trace token cost: approximately five hundred tokens. This is the highest-ROI token expenditure in the entire system. Each counter-trace costs sixty to one hundred tokens and provides targeted immunization against the specific failure mode most likely to occur at the corresponding regulatory intersection.

6.4 The Composition Grammar

Primitives compose according to rules determined by the intersection signature computed by Layer 1. The grammar is a small, explicit decision tree—not a neural network, not a graph traversal algorithm—that takes the intersection signature as input and outputs the primitive chain.

Single-authority queries: $T \rightarrow S \rightarrow M \rightarrow B \rightarrow K$

Two-authority queries (edge): $T(a_1) \rightarrow T(a_2) \rightarrow H(a_1,a_2) \rightarrow S \rightarrow M \rightarrow B \rightarrow K$

Two-authority with conflict: $T(a_1) \rightarrow T(a_2) \rightarrow C(a_1,a_2) \rightarrow H \rightarrow S \rightarrow M \rightarrow B \rightarrow K$

Three or more authorities (full-mesh): [$T(a_n)$ for all] → [$C(a_i,a_j)$ for conflicting pairs] → [H for each conflict] → G → S → M → B → K

A compliance officer can read this grammar and verify it in an afternoon. This auditability is not incidental—it is a primary design goal. If the reasoning methodology cannot be understood and validated by the humans accountable for regulatory compliance, it should not be deployed.

6.5 Coverage and Scalability

With approximately sixty to eighty validated primitives (eight types, each with eight to twelve exemplars covering major authority combinations), the grammar generates correct reasoning chains for any authority intersection—including intersections that have never been encountered. If a new regulatory development creates a previously nonexistent intersection (for example, a novel FDA regulation intersecting with OPTN allocation policy), the grammar composes the relevant HIERARCHY_RESOLVE and CONFLICT_ADJUDICATE primitives for the new pair from existing exemplars. No new trace generation is required for the reasoning pattern; only the factual content (retrieved via RAG) changes.

This represents a fundamental improvement over approaches that store complete reasoning demonstrations. A library of five hundred complete demonstrations covers five hundred scenarios.

A grammar of sixty primitives generates correct reasoning chains for a combinatorial space of intersections that vastly exceeds what any static library can enumerate.

7. Layer 3: The Safety Automaton

The automaton is a formal specification of the minimum viable reasoning process for organ procurement regulatory questions. It serves three functions: structural safety enforcement, completeness verification, and non-technical auditability.

7.1 The State Machine

The regulatory reasoning automaton defines the following states and transitions. Every query must traverse these states in order; no state may be skipped.

> INTAKE → AUTHORITY_ROUTE (Layer 1 output) → TEMPORAL_CLASSIFY (for each authority) → [If multi-authority: CONFLICT_TEST → HIERARCHY_RESOLVE; if irreconcilable: GAP_DETECT] → SYNTHESIZE + CALIBRATE → CASCADE_MAP → BOUNDARY_ENFORCE → OUTPUT

Two transitions warrant emphasis. First, GAP_DETECT is triggered when hierarchy resolution produces ambiguity rather than a clear determination—the system recognizes that the regulatory environment itself does not provide a clear answer. Second, BOUNDARY_ENFORCE is positioned *after* CASCADE_MAP because the cascade itself may trigger boundary-adjacent obligations that require checking.

7.2 The Structural Safety Guarantee

There is no path from any state to OUTPUT that bypasses BOUNDARY_ENFORCE. This is not a policy instruction. It is a structural property of the automaton's topology. The Human Line—the set of decisions that must remain with human practitioners regardless of AI capability—is enforced by the absence of a graph edge, not by a textual admonition.

The practical consequence is significant. No amount of prompt injection, adversarial framing, or escalating scope creep can cause the system to produce a clinical determination, an eligibility assessment, or a family communication recommendation. The path through which such output would be generated does not exist in the automaton. The model's creativity operates within structural constraints that it cannot override because they are properties of the architecture, not instructions it must choose to follow.

7.3 Completeness Verification

Given any model output, the automaton enables mechanical verification. Did the model visit TEMPORAL_CLASSIFY for every authority in the itinerary? Did it visit BOUNDARY_ENFORCE before producing output? Did it visit GAP_DETECT when conflict resolution was ambiguous? This verification can be performed by a lightweight post-processing step: a script that parses the model's output for state markers and flags any deviation from the required path. Human review is reserved for content correctness; process completeness is verified mechanically.

If the model's output fails completeness verification—missing required state markers or visiting states out of order—the system either re-prompts the model with a correction instruction or returns a DEFERRED output with escalation guidance. It does not present an incomplete reasoning chain as a determination.

7.4 Auditability for Non-Technical Stakeholders

A CMS surveyor, compliance officer, or board member does not need to understand large language models. They need to understand the state diagram: does this process visit all the steps a human regulatory expert would visit? Are there any paths that skip the boundary check? If the answer is satisfactory, the AI system's reasoning *process* is validated—regardless of which language model is running underneath. This collapses the AI trust problem into a finite, auditable structure that maps to existing compliance review processes.

8. Layer 4: The Inference Engine

At inference time, the preceding layers compose into a pipeline that assembles a context window payload, submits it to the language model, and validates the output.

8.1 Triple Retrieval

The inference engine performs three parallel retrievals, each serving a distinct function:

Policy chunk retrieval (standard RAG). Authoritative regulatory text for the authorities identified in the itinerary. This is the factual substrate—the ground truth that governs the answer's content.

Primitive retrieval (reasoning RAG). The composition grammar selects the primitive chain based on the intersection signature. The retrieval engine fetches the skeletal forms (always) and expanded forms (when complexity warrants) for the specified primitives. For calibrated uncertainty, two to three variant exemplars of the same primitive type are retrieved; convergence signals confidence while divergence signals that the intersection may require escalation.

Obligation context retrieval. From the case's running obligation state (Layer 7), the system retrieves what has already been determined or documented for this case and what remains outstanding. This prevents re-derivation of previously resolved questions and enables the model to factor prior case decisions into current reasoning.

8.2 Token Budget

Component	Tokens	Notes
System prompt (automaton, grammar, gold primitives, counter-traces, output contract, Human Line)	4,000–5,500	Permanent, present on every query
Authority itinerary (from Layer 1)	200–600	Includes temporal phase model
Retrieved policy chunks (RAG)	3,000–6,000	Standard RAG retrieval
Retrieved primitives (2–4 skeletal or expanded)	400–1,600	Grammar-compiled, calibrated to coordinator expertise
Counter-traces (2–3 relevant)	120–300	Targeted to query risk profile
Obligation context (from case state)	200–800	Running case determinations
Query + response space	2,000–4,000	User input + model output
Total	~10,000–19,000	Less than 10% of 200K context window

The system prompt of four to five thousand tokens is dramatically smaller than approaches that embed complete reasoning demonstrations (which typically require twelve to twenty-four thousand tokens for comparable coverage), while providing broader coverage because the grammar generates reasoning chains combinatorially from the primitives.

8.3 Primitive Divergence for Calibrated Uncertainty

For edge and full-mesh queries, the retriever pulls two to three primitives of the same type that approach the problem from slightly different angles. The model is instructed to note whether the primitives converge or diverge:

Convergence (all primitives agree on authority identification, hierarchy, and conclusion): high confidence. Multiple reasoning approaches reach the same determination.

Partial divergence (primitives agree on authority identification but differ on resolution): moderate confidence. The specific point of divergence is flagged and both paths are presented.

Full divergence (primitives disagree on which authorities are relevant): low confidence. The query represents a genuinely novel intersection. The system recommends supervisor review and flags the query as a candidate for new primitive generation.

This is ensemble reasoning at the methodology level—not asking multiple models for opinions, but showing the reasoning engine multiple valid approaches and measuring agreement. The cost is approximately three hundred extra tokens for additional exemplars. The gain is calibrated uncertainty that maps to clinical decision support norms, where practitioners are trained to calibrate trust based on evidence convergence.

8.4 Coordinator Expertise Calibration

The system maintains a lightweight coordinator profile (set at login or inferred from role data) that calibrates output detail:

Profile	Primitive Form	Counter-Trace Inclusion	Evidence Pack Detail
Expert (10+ years)	Skeletal only	Novel intersections only	Abbreviated
Experienced (3–10 years)	Skeletal default, expanded on request	Standard set	Standard
Developing (0–3 years)	Expanded default	Full set with pedagogical notes	Full with annotations

The expert coordinator is not buried in explanations she has internalized. The developing coordinator receives the scaffolding she needs. This requires no machine learning—merely a profile flag and conditional retrieval rules.

9. Layer 5: The Output Contract

Every response follows a fixed output structure. This structure is non-negotiable—it is what makes the system audit-grade for CMS surveyors and defensible in any review context.

9.1 The Nine-Field Output Schema

1. Regulatory State Anchor. Case date and time assumed; applicable regulatory phase (e.g., “CMS-3409-P transition period”). If the date or phase is incorrect, results may change.

2. Authority Itinerary. Ordered list of authorities consulted, with status and scope for each. Precedence statement.

3. Determination. What the system can safely state, in clear operational language. Or, if the system cannot make a determination: DEFERRED, with explanation.

4. Regulatory Cascade. Downstream obligations triggered by this determination, enumerated by authority, with deadlines where applicable. These obligations are automatically added to the case obligation tracker (Layer 7).

5. Performance Metric Impact. CMS-3409-P metrics affected by this decision, direction of impact, and documentation required. Explicit caveat: "This is informational. Clinical and operational decisions should not be driven by metric optimization."

6. Confidence. HIGH, MODERATE, or LOW, with explicit justification. If primitive divergence was detected, both paths are presented with the specific point of divergence identified.

7. Boundaries Respected. Explicit enumeration of what the response does NOT determine: clinical viability, transplant center decision-making, family communication, or other Human Line categories.

8. Escalation Path. When applicable: who to contact, when, why. Specific routing (QAPI, legal, medical director, ethics committee) based on the nature of the uncertainty or gap.

9. Evidence Pack. Regulatory citations (specific provisions), primitive IDs used, counter-traces checked, automaton state path traversed, itinerary hash for snapshot verification.

9.2 Operational Value

The Regulatory Cascade field makes downstream obligations visible at the moment of decision. The coordinator does not discover three days later that she missed a reporting requirement triggered by a 3 AM decision—the system tells her when the decision is made. The Performance Metric Impact field addresses the CMS-3409-P reality directly: every procurement decision has a metric consequence, and coordinators need operational awareness of that consequence even though metrics should not drive clinical or operational choices.

The Evidence Pack serves a legal and compliance protective function. If a regulatory question is later challenged, the OPO can point to a documented reasoning chain showing which authorities were consulted, in what order, how conflicts were resolved, what boundaries were respected, and what the escalation path was. The automaton path within the Evidence Pack provides proof that the reasoning process was structurally complete.

10. Layer 6: Continuous Evolution

The reasoning infrastructure must improve continuously through operational use, adapt to regulatory changes, and propagate improvements across the OPO community. Layer 6 achieves this through three mechanisms: structured feedback, harm-prioritized generation, and surgical maintenance.

10.1 Structured Feedback Logging

Coordinator interactions produce structured signals:

Action	System Response
Accepts the response	Boost retrieval weight of primitives used
Corrects the response	Correction + context enters generation pipeline at cross-verification stage; if correction reveals primitive error, flag for regeneration; if it reveals a gap, flag in the grammar
Escalates to supervisor	Query exceeded current coverage; auto-generate scenario; enters pipeline at scenario generation; prioritized by Expected Harm Score
Reports wrong authority set	Gold signal for router improvement; logged for deterministic router refinement

No machine learning infrastructure is required. This is structured logging feeding a prioritized queue for human-reviewed primitive generation.

10.2 Expected Harm Scoring

The offline generation pipeline (expensive, performed with extended test-time compute) should not target coverage uniformity. It should target *expected harm reduction.* The Expected Harm Score for a query class is defined as:

$$\text{EHS} = \text{Frequency} \times \text{Consequence Severity} \times \text{Observed Uncertainty}$$

Where Frequency is measured from production logs, Consequence Severity is scored by domain experts (regulatory citation risk, organ loss risk, time-criticality), and Observed Uncertainty is measured by primitive divergence rate and escalation rate in production. The generation budget is allocated to query classes with the highest EHS. Full-mesh queries are rare but carry disproportionate consequence severity; the EHS naturally prioritizes them without manual intervention.

10.3 Regulatory Diff Propagation

When a regulatory change occurs, the system computes a structured diff and triages affected primitives:

Category	Description	Action	Cost
Temporal-only	Provision changed status (proposed → final); reasoning pattern unchanged	Auto-patch temporal metadata	Seconds
Content change	Provision text changed; reasoning path preserved	Auto-patch cited text; light human review	Minutes
Pattern affected	Change alters hierarchy, creates new intersection, or modifies precedence	Full regeneration via generation pipeline	Hours per primitive

For a typical regulatory event (e.g., CMS finalizing a proposed rule), approximately eighty percent of affected primitives are temporal-only (auto-patched), fifteen percent are content changes (light review), and five percent require full regeneration. The expensive pipeline runs on a handful of primitives instead of regenerating hundreds of complete demonstrations. This represents approximately a ninety-five percent reduction in staleness resolution cost.

10.4 Regulatory Horizon Scanning

The evolution layer should not only react to regulatory changes; it should anticipate them. A lightweight monitoring process (RSS feeds from the Federal Register and OPTN website, reviewed weekly by a domain expert) maintains a regulatory horizon of expected changes. When a pending OPTN policy amendment is expected to affect specific primitives, alternative versions are drafted in advance. When the amendment finalizes, the prepared primitive deploys immediately—no generation delay, no staleness window.

10.5 Automaton Path Analytics

The runtime path logs from Layer 3 feed back into the evolution pipeline. Paths that consistently produce accepted responses indicate working primitives. Paths that consistently produce corrections indicate primitive errors. Paths that consistently produce escalations indicate coverage gaps. Paths that have never been traversed indicate either genuinely rare intersections or router misclassification. This closes the loop between how the system reasons and how well it reasons, making the automaton not just a safety structure but an observability framework.

11. Layer 7: Proactive Regulatory Navigation

This layer transforms the architecture from a regulatory question-answering system into a regulatory co-pilot. It addresses the highest-value failure mode in organ procurement: not the wrong answer, but the *unasked question*—the regulatory obligation the coordinator did not know existed for this case.

11.1 The Obligation Tracker

For each active case, the system maintains a running state of regulatory obligations—what has been fulfilled, what is pending, and what is upcoming based on the case's current stage and decisions made so far. The tracker is a structured data store (JSON, SQLite, or integrated into the case management system) maintained by deterministic logic, not language model inference.

The tracker is populated from three sources. First, **case-type templates**: when a case opens, a template of standard obligations for that case type pre-populates (DCD cases have different obligation sets than DBD cases). These templates are hand-built by domain experts, versioned with the regulatory corpus, and maintained alongside the authority router decision structure. Second, **CASCADE_MAP outputs**: every time the system produces a determination with a regulatory cascade (Layer 5, field 4), the cascaded obligations are added to the tracker. Third, **stage transitions**: as the case moves through stages, stage-specific obligations activate.

11.2 Proactive Alerts

The obligation tracker enables three categories of proactive communication that do not require the coordinator to ask a question:

Deadline alerts. "Organ offer for [organ] has been pending for [X hours]. OPTN Policy 5.6.B requires response within [Y hours]. [Z hours] remaining."

Stage-transition briefings. When a case moves from authorization to allocation: "This case is entering allocation. Three regulatory obligations activate at this stage. Key intersection to be aware of: this DCD case with an out-of-territory donor involves a UAGA↔OPTN jurisdictional question at allocation. If you have questions about this intersection, ask me."

Unfulfilled obligation warnings. "This case has been in the allocation stage for [X hours]. The following obligation remains unfulfilled: [obligation]. This may affect CMS-3409-P performance metric documentation. Recommended action: [specific step]."

11.3 Decision Consequence Modeling

When a coordinator faces a decision point, the system models the downstream regulatory consequences of different choices—before the decision is made.

Example: The transplant center requests extended cold ischemia time.

Path A: Approve extended cold ischemia time. Triggers OPTN documentation requirement for extended preservation rationale. Triggers CMS QAPI documentation for variance from standard protocol. CMS-3409-P metric impact: organ utilization rate maintained; potential documentation audit flag for preservation variance.

Path B: Decline extended cold ischemia and re-offer organ. Triggers OPTN offer decline documentation. Triggers re-entry into allocation sequence with associated timeline obligations. CMS-3409-P metric impact: offer acceptance rate affected; re-allocation timeline begins.

> "Both paths are regulatory compliant. The choice is clinical and operational, not regulatory. This analysis covers only the regulatory dimensions."

The coordinator sees the full regulatory landscape of her decision before she makes it. She does not discover missing documentation requirements after the fact. She does not learn about metric impacts during the next CMS survey. The regulatory consequences are visible at the decision point, in real time, during the case.

11.4 Implementation Architecture

Layer 7 is predominantly deterministic. The language model fires only when natural language generation is needed (stage briefings, consequence modeling prose). The obligation tracker, deadline calculations, and template-driven pre-population are all programmatic. This keeps compute cost low and behavior predictable.

At case close, the system generates a summary of all obligations fulfilled, flagging any gaps for QAPI review. This summary integrates directly with the OPO's quality assurance workflow, creating a closed loop between AI-assisted decision support and organizational quality improvement.

12. The Generation Pipeline

Reasoning primitives are generated offline using extended test-time compute and validated through a multi-stage pipeline designed to maximize quality while managing cost.

12.1 Scenario Generation

A frontier language model, provided with the full regulatory corpus, generates a comprehensive set of regulatory scenarios targeting each cell in the combinatorial matrix of authority intersections. Scenarios include realistic operational situations drawn from common OPO workflow patterns and deliberately constructed edge cases representing plausible but unprecedented regulatory intersections. Scenario generation targets *primitive exemplars*, not complete demonstrations: for each primitive type, scenarios exercise the primitive across different authority combinations and operational contexts.

12.2 Multi-Agent Debate

For edge and full-mesh primitives, the generation stage employs structured debate among three specialized agents rather than relying on a single model perspective:

The Compliance Advocate optimizes for CMS and regulatory risk minimization.

The Utilization Advocate optimizes for organ utilization and allocation efficiency.

The Boundary Guardian detects and penalizes Human Line violations.

The agents debate the scenario until convergence. The synthesized debate log—how the Compliance Advocate's reasoning was bounded by the Guardian, how the Utilization Advocate's interpretation was constrained by the Compliance Advocate—becomes the reasoning chain in the primitive. A single model has a unified perspective; forcing specialized agents into structured conflict produces reasoning chains that have been pressure-tested from multiple angles before they enter the library.

12.3 Cross-Verification and Counter-Trace Generation

Each primitive undergoes adversarial review by a separate model instance. The reviewer is given the scenario and the reasoning chain and tasked with identifying errors, missed authorities, overconfident claims, or boundary violations. This stage also generates the paired counter-trace: "What is the most plausible wrong reasoning for this scenario?" The compressed failure demonstration and forensic diagnosis are stored alongside the primitive.

12.4 Human Curation

A domain expert reviews cross-verified primitives, corrects remaining errors, validates the composition grammar (verifying that primitive chains produce correct reasoning for test scenarios), and selects the gold primitives for the system prompt. The gold primitives are selected for *pattern diversity* over topic coverage: one exemplar per primitive type, chosen for pedagogical clarity.

12.5 Cost Amortization

Extended test-time compute is expensive per generation but the cost amortizes. A fifteen-minute deep-think run generating a full-mesh primitive may cost one to two dollars. If that primitive subsequently improves inference quality across ten thousand future queries at standard latency, the effective cost is fractions of a cent per query. The generation pipeline is an investment in reasoning infrastructure whose returns scale with deployment volume.

13. Community Architecture: The Archetypal Abstraction

The fifty-six OPOs operating in the United States share a common regulatory core—all must comply with the same CMS conditions of participation, OPTN policies, and NOTA provisions—but differ in internal procedures, territorial considerations, staffing models, and state-specific UAGA requirements.

13.1 Layer A and Layer B

Layer A (Archetypal, portable, public) contains the universal primitive library, the composition grammar, the counter-trace catalog, and the automaton specification. These are generated against an abstracted "common core" OPO—the set of policies and reasoning patterns shared across all OPOs by regulatory mandate. Any OPO can adopt Layer A immediately.

Layer B (Organization-specific, private) contains primitives generated against a particular OPO's internal SOPs, territorial boundaries, staffing model, state UAGA, and CMS-3409-P performance tier position. Each OPO generates its own Layer B.

At inference time, the retrieval layer draws from both layers: Layer A provides the reasoning methodology for how any OPO navigates the relevant regulatory intersection, and Layer B provides organization-specific details that particularize the response.

13.2 AOPO as Custodian

The Association of Organ Procurement Organizations is the natural custodian of the Layer A primitive library. This aligns with AOPO's existing role as a professional standards body and gives the association a concrete, high-value function in the AI era: curating community contributions, maintaining the composition grammar, publishing the counter-trace catalog, and coordinating primitive regeneration when federal regulations change.

The practical mechanism is a shared repository with a standard contribution workflow. When one OPO discovers a failure mode and generates a counter-trace, that counter-trace becomes available to all fifty-six organizations. When another OPO encounters a novel full-mesh intersection and generates new primitives, those primitives enter the community library. Every organization's operational experience strengthens every other organization's reasoning scaffold.

This is herd immunity for regulatory reasoning. And it aligns with the cooperative norms of the organ procurement community, where the shared mission of saving lives takes precedence over organizational competition.

14. Discussion

14.1 Portability

Because the architecture operates through the context window rather than through weight modification, it is portable across any language model that accepts system prompts. A primitive library developed and validated against one model can be deployed without modification on another. This eliminates vendor lock-in for the reasoning component—the most intellectually valuable and expensive-to-produce asset. If an organization changes model providers, the reasoning infrastructure and its embedded domain expertise transfer fully.

14.2 Relationship to Fine-Tuning

This architecture is complementary to, not a replacement for, fine-tuning. Fine-tuning embeds patterns into weights, enabling behavioral modification without consuming context window tokens. Context-window deployment consumes tokens but requires no weight access. For organizations using proprietary frontier models, context-window deployment is the only available mechanism for domain reasoning transfer. For organizations using open-weight local models, the primitive library can serve dual purpose: deployed as context-window primitives for immediate use, and converted to supervised fine-tuning format for weight embedding. The primitive's scenario field becomes the user turn; the reasoning chain becomes the assistant turn.

14.3 Cost Structure

The framework's cost structure is front-loaded and amortizing. The initial investment—building the primitive library (sixty to eighty primitives via the generation pipeline), the authority router, and the automaton—requires several weeks of focused effort. The ongoing cost is near-zero per query (standard language model inference) with periodic maintenance costs triggered by regulatory changes (largely automated via diff propagation) and community contributions (lightweight curation). For OPOs facing decertification under CMS-3409-P, the relevant comparison is not the cost of the infrastructure but the cost of regulatory error.

14.4 Applicability Beyond OPO

While this paper focuses on the OPO regulatory domain, the architecture is applicable to any bounded, multi-authority regulatory environment. Examples include clinical trial compliance (FDA regulations intersecting with IRB requirements and institutional policies), healthcare billing (CMS billing rules intersecting with payer-specific requirements), and environmental permitting (federal EPA regulations intersecting with state agencies and local ordinances). The key enabler is domain boundedness: the set of authoritative sources must be finite and enumerable so that the primitive library and composition grammar are tractable. The specific primitives are domain-dependent; the architecture is domain-agnostic.

15. Limitations

Several limitations constrain the current framework. First, the architecture has not yet been empirically validated through controlled experimentation comparing model performance with and without the compositional reasoning infrastructure. The framework is presented as an architectural design; empirical validation is planned as immediate future work.

Second, the risk of over-anchoring—where the model treats primitive reasoning as ground truth rather than methodological guidance—has been identified but not quantified. The system prompt includes override directives (retrieved regulatory text governs facts; primitives demonstrate methodology), but the effectiveness of these directives under edge conditions requires systematic testing.

Third, the deterministic authority router assumes that the mapping from query features to authority sets can be maintained by hand as a decision structure. For the OPO domain with its finite authority set, this is tractable. For domains with larger or more fragmented authority structures, the router may require more sophisticated classification approaches.

Fourth, primitive quality is bounded by the capabilities of the generating models and the expertise of human curators. Errors that survive the multi-agent debate, cross-verification, and human review would propagate as authoritative methodology. The multi-stage pipeline mitigates but does not eliminate this risk.

Fifth, the proactive navigation layer (Layer 7) depends on integration with the OPO's case management system. The depth and quality of proactive capabilities are constrained by the data available from that integration. Organizations with mature, well-structured case management systems will realize greater value from Layer 7 than those with fragmented or manual tracking processes.

Sixth, attention dynamics in large context windows may reduce the model's effective uptake of primitives as total context size increases. The skeletal-versus-expanded form strategy mitigates this, but optimal primitive positioning, count, and length within the context window are empirical questions requiring investigation.

16. Conclusion

The organ procurement community faces a convergence of regulatory pressure and technological opportunity. CMS-3409-P creates accountability without providing the capability infrastructure to meet it. AI systems offer transformative potential but require domain-specific reasoning alignment that retrieval-augmented generation alone does not provide.

This paper presents a compositional architecture that treats regulatory reasoning as infrastructure—manufactured, versioned, portable, and continuously improved. By decomposing multi-authority reasoning into typed primitives organized by a formal grammar, routing authority identification deterministically, enforcing safety boundaries through automaton topology, and extending beyond reactive question-answering to proactive regulatory co-navigation, the framework provides a practical, portable, and economically viable path forward for all fifty-six OPOs.

The system prompt is approximately five thousand tokens. The compositional grammar covers the full space of regulatory intersections from a library of sixty to eighty primitives. The safety guarantee is structural: the Human Line is enforced by the topology of the automaton, not by textual instruction. The output contract produces audit-grade evidence packs. The evolution loop improves the infrastructure continuously through operational feedback and community contribution. The proactive navigation layer surfaces regulatory obligations before they are missed.

We release all specifications under open license and invite the OPO community to participate in building the Layer A archetypal primitive library as a shared resource. The Association of Organ Procurement Organizations is the natural custodian of this community asset.

The architecture is built on a thesis that extends beyond any single regulatory domain: **reasoning competence is separable, storable, and transferable.** It can be manufactured once by the best available intelligence, validated by human experts, and deployed infinitely at near-zero marginal cost. The OPO community is the proving ground, but the principle applies wherever bounded regulatory environments impose complex, multi-authority reasoning requirements on human practitioners.

Every primitive composed is an intersection mastered. Every counter-trace is a failure mode eliminated. Every evidence pack is a decision that can withstand scrutiny. Every obligation tracked is a requirement that will not be missed.

Every organ saved is a life continued.